

I am going to go with the idea to do a trivia style game. My thinking for this trivia game is to build it around a dictionary of categories, where each category name (like Science, History, Sports) maps to a list of questions. I picked a dictionary specifically because the whole point is looking something up by name, which is exactly what dictionaries are good at, so that gives me a real reason for the design instead of just shoving one in. Each individual question is going to be its own little dictionary too, with the question text, a list of multiple choice options, and the correct answer, so I end up with a dictionary inside a list inside a dictionary, which shows I actually understand nesting and not just the basics.

When I show a question I'll loop through the options list with a counter so they print out as 1, 2, 3, 4, and the player types a number to answer. I'm going to validate that input hard so if they type something that isn't one of the valid choices (like "5") the program just reasks instead of crashing, which I'll handle with a while loop that only breaks once the input is actually valid.

For the gameplay itself the player gets 3 lives and loses one every time they miss a question and the round ends early if they hit zero lives which is a nice use of a flag pattern with a while loop. I also want a scoring system with a bonus twist so that instead of every right answer being worth the same I'll track a streak and if the player gets several correct in a row they earn bonus points on top of the base points. That gives me a natural reason to use nested if/elif logic and it makes the game more fun because momentum actually matters.

Finally, I'll wrap the whole thing in an outer while loop so after a category finishes the player can pick another one or quit and I'll keep running totals (total correct, total asked, total score) in a session stats dictionary so I can print a real summary report at the end with their percentage.

Overall this hits strings and numbers, lists, dictionaries and nesting, if/elif/else, for loops, while loops and user input all in a way that actually makes sense for the project rather than feeling forced.